

Processando config 1, seed 2, teste: Rabbit
 xx
 Starting Rabbit: nb = 16777216
 Version: TestU01 1.2.3
 xx

 HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
 Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

smultin_MultinomialBitsOver test:

 N = 2, n = 8388608, r = 0, s = 32, L = 38,
 Sparse = TRUE

Number of bits = n = 8388608
 Number of cells = 2^L = 274877906944
 Expected number per cell = $1 / 32768$
 $EColl = n^2 / (2k) = 128$
 Hashing = TRUE

Collision test
 CollisionOver: density = $n / k = 1 / 32768.00$
 Expected number of collisions = $\mu = 128.00$

 Results of CollisionOver test:

POISSON approximation :
 Expected number of collisions = $N \cdot \mu$: 256.00
 Observed number of collisions : 297
 p-value of test : 6.6e-3

 Total number of cells containing j balls

j = 0	:	549739036969
j = 1	:	16776622
j = 2	:	297
j = 3	:	0
j = 4	:	0
j = 5	:	0

 CPU time used : 00:00:04.56

Generator state:
 16777216 bits have been read.

 HOST = matheus-IdeaPad-3-15ALC6, Linux

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snpair_ClosePairsBitMatch test:

N = 1, n = 262144, r = 0, t = 2

Max of all bit distances : 18.00
p-value of test : 0.12

CPU time used : 00:00:00.04

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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snpair_ClosePairsBitMatch test:

N = 1, n = 131072, r = 0, t = 4

Max of all bit distances : 12.00
p-value of test : 1.9e-6 *****

CPU time used : 00:00:00.07

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

svaria_AppearanceSpacings test:

N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of $n = (K + Q)L = 15728640$ bits
Q = 524288 initialization blocks
K = 524288 blocks for the test
the blocks have L = 15 bits

Normal statistic : 0.12
p-value of test : 0.45

CPU time used : 00:00:00.00

Generator state:
67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LinearComp test:

N = 1, n = 18176, r = 0, s = 32

Number of degrees of freedom : 7
Chi2 statistic for size of jumps : 8.61
p-value of test : 0.28

Normal statistic for number of jumps : -51.95
p-value of test : 1 - eps1 *****

CPU time used : 00:00:00.06

Generator state:
67127040 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
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scomp_LempelZiv test:

N = 1, n = 16777216, r = 0, s = 32, k = 24

Normal statistic : -0.32
p-value of test : 0.62

CPU time used : 00:00:00.39

Generator state:

83904256 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sspectral_Fourier1 test:

N = 1, n = 1048576, r = 0, s = 32, k = 20

Normal statistic : 0.16
p-value of test : 0.44

CPU time used : 00:00:00.21

Generator state:

84952832 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sspectral_Fourier3 test:

N = 8065, n = 2048, r = 0, s = 32, k = 11

Kolmogorov-Smirnov+ statistic = D+ : 0.036
p-value of test : 0.26

Kolmogorov-Smirnov- statistic = D- : 0.036
p-value of test : 0.26

Anderson-Darling statistic = A2 : 1.16
p-value of test : 0.29

CPU time used : 00:00:00.11

Generator state:

101728032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin
```

sstring_LongestHeadRun test:

N = 1, n = 60, r = 0, s = 32, L = 279616

Number of degrees of freedom : 3
Chi-square statistic : 2.83
p-value of test : 0.42

Global longest run of 1 : 24.00
p-value of test : 0.39

CPU time used : 00:00:00.06

Generator state:
118504992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin
```

sstring_PeriodsInStrings test:

N = 1, n = 524288, r = 0, s = 31

Number of degrees of freedom : 94
Chi-square statistic : 88.19
p-value of test : 0.65

CPU time used : 00:00:00.03

Generator state:
135282208 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
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```

sstring_HammingWeight test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Number of degrees of freedom : 22
Chi-square statistic : 20.72
p-value of test : 0.54

CPU time used : 00:00:00.00

Generator state:
152059424 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.68
p-value of test : 0.25

CPU time used : 00:00:00.00

Generator state:
168836640 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_HammingCorr test:

N = 1, n = 262144, r = 0, s = 32, L = 64

Normal statistic : -0.72
p-value of test : 0.76

CPU time used : 00:00:00.00

Generator state:
185613856 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_HammingCorr test:

N = 1, n = 131072, r = 0, s = 32, L = 128

Normal statistic : 0.049
p-value of test : 0.48

CPU time used : 00:00:00.00

Generator state:
202391072 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 185.57
p-value of test : 0.31

CPU time used : 00:00:00.00

Generator state:
219168288 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 297.97
p-value of test : 0.60

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_HammingIndep test:

N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 489
Chi-square statistic : 2244.04
p-value of test : eps *****

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

sstring_AutoCor test:

N = 1, n = 16777153, r = 0, s = 32, d = 1


```
-----
Normal statistic          :    0.92
p-value of test          :    0.18
-----
```

```
CPU time used            : 00:00:00.00
```

```
Generator state:
269499904 bits have been read.
```

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*****
```

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HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin
```

```
sstring_AutoCor test:
```

```
-----
N = 1, n = 16777154, r = 0, s = 32, d = 2
-----
```

```
-----
Normal statistic          :   -0.67
p-value of test          :    0.75
-----
```

```
CPU time used            : 00:00:00.02
```

```
Generator state:
286277088 bits have been read.
```

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*****
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```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin
```

```
sstring_Run test:
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```
-----
N = 1, n = 3355443, r = 0, s = 32
-----
```

```
-----
Total number of 1 runs: 3355447
```

```
Number of degrees of freedom : 36
Chi2 statistic for number of runs : 38.23
p-value of test : 0.37
-----
```

```
-----
Total number of bits: 13421088
```

```
Normal statistic for number of bits : -0.13
p-value of test : 0.55
```

CPU time used : 00:00:00.04

Generator state:
299698176 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

smarsa_MatrixRank test:

N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

Number of degrees of freedom : 3
Chi-square statistic : 267.45
p-value of test : eps *****

CPU time used : 00:00:00.06

Generator state:
316475392 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

smarsa_MatrixRank test:

N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

Number of degrees of freedom : 2
Chi-square statistic : 1056.73
p-value of test : eps *****

CPU time used : 00:00:00.03

Generator state:
333166592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

swalk_RandomWalk1 test:

N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

Test on the values of the Statistic H

Number of degrees of freedom	:	42	
ChiSquare statistic	:	1065.76	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	42	
ChiSquare statistic	:	418.34	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	64	
ChiSquare statistic	:	103.26	
p-value of test	:	1.4e-3	

Test on the values of the Statistic R

Number of degrees of freedom	:	36	
ChiSquare statistic	:	25.38	
p-value of test	:	0.91	

Test on the values of the Statistic C

Number of degrees of freedom	:	22	
ChiSquare statistic	:	18.11	
p-value of test	:	0.70	

CPU time used : 00:00:00.06

Generator state:
349943808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

swalk_RandomWalk1 test:

N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

Test on the values of the Statistic H

Number of degrees of freedom	:	94	
ChiSquare statistic	:	382.14	
p-value of test	:	eps	*****

Test on the values of the Statistic M

Number of degrees of freedom	:	94	
ChiSquare statistic	:	441.38	
p-value of test	:	eps	*****

Test on the values of the Statistic J

Number of degrees of freedom	:	512	
ChiSquare statistic	:	546.00	
p-value of test	:	0.14	

Test on the values of the Statistic R

Number of degrees of freedom	:	90	
ChiSquare statistic	:	99.58	
p-value of test	:	0.23	

Test on the values of the Statistic C

Number of degrees of freedom	:	50	
ChiSquare statistic	:	66.04	
p-value of test	:	0.06	

CPU time used : 00:00:00.03

Generator state:

366721024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config1_seed2.bin

swalk_RandomWalk1 test:

N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

Test on the values of the Statistic H

Number of degrees of freedom	:	130
ChiSquare statistic	:	140.21
p-value of test	:	0.26

Test on the values of the Statistic M

Number of degrees of freedom	:	131
ChiSquare statistic	:	179.29
p-value of test	:	3.3e-3

Test on the values of the Statistic J

Number of degrees of freedom	:	161
ChiSquare statistic	:	140.75
p-value of test	:	0.87

Test on the values of the Statistic R

Number of degrees of freedom	:	130
ChiSquare statistic	:	113.80
p-value of test	:	0.84

Test on the values of the Statistic C

Number of degrees of freedom	:	89
ChiSquare statistic	:	96.49
p-value of test	:	0.28

CPU time used : 00:00:00.03

Generator state:

383497824 bits have been read.

===== Summary results of Rabbit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin
Number of bits:	16777216
Number of statistics:	39

Total CPU time: 00:00:06.00
 The following tests gave p-values outside [0.001, 0.9990]:
 (eps means a value < 1.0e-300):
 (eps1 means a value < 1.0e-15):

	Test	p-value
3	ClosePairsBitMatch, t = 4	1.9e-6
5	LinearComp	1 - eps1
17	HammingIndep, L = 64	eps
21	MatrixRank, 32 x 32	eps
22	MatrixRank, 320 x 320	eps
24	RandomWalk1 H	eps
24	RandomWalk1 M	eps
25	RandomWalk1 H (L = 1024)	eps
25	RandomWalk1 M (L = 1024)	eps

 All other tests were passed